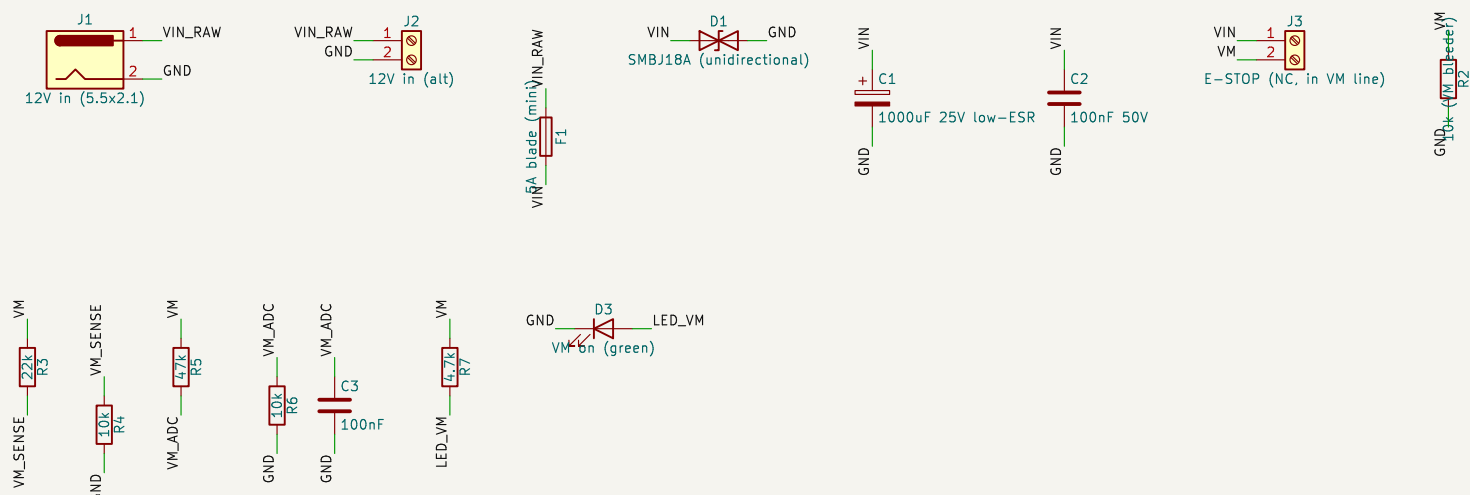
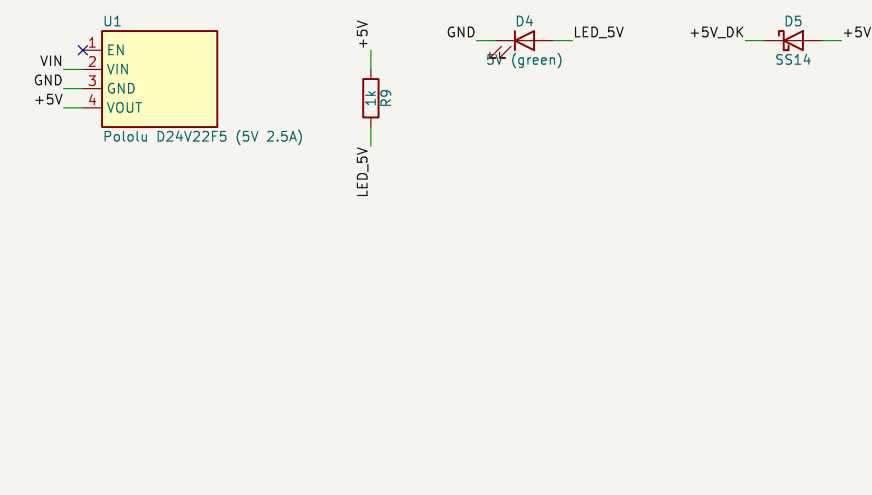


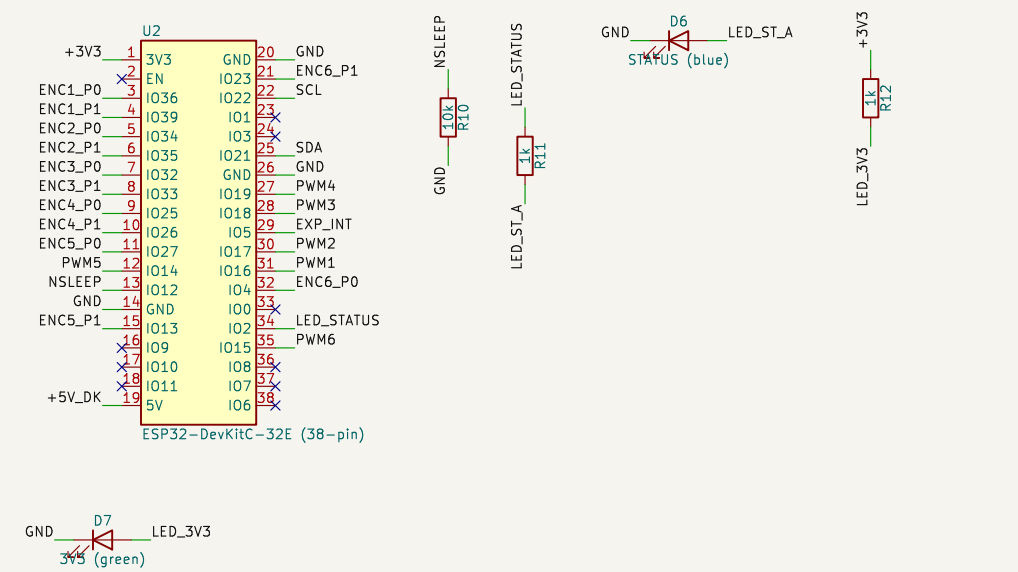
POWER IN, REVERSE PROTECTION, E-STOP (no relay: NC contacts in VM line)



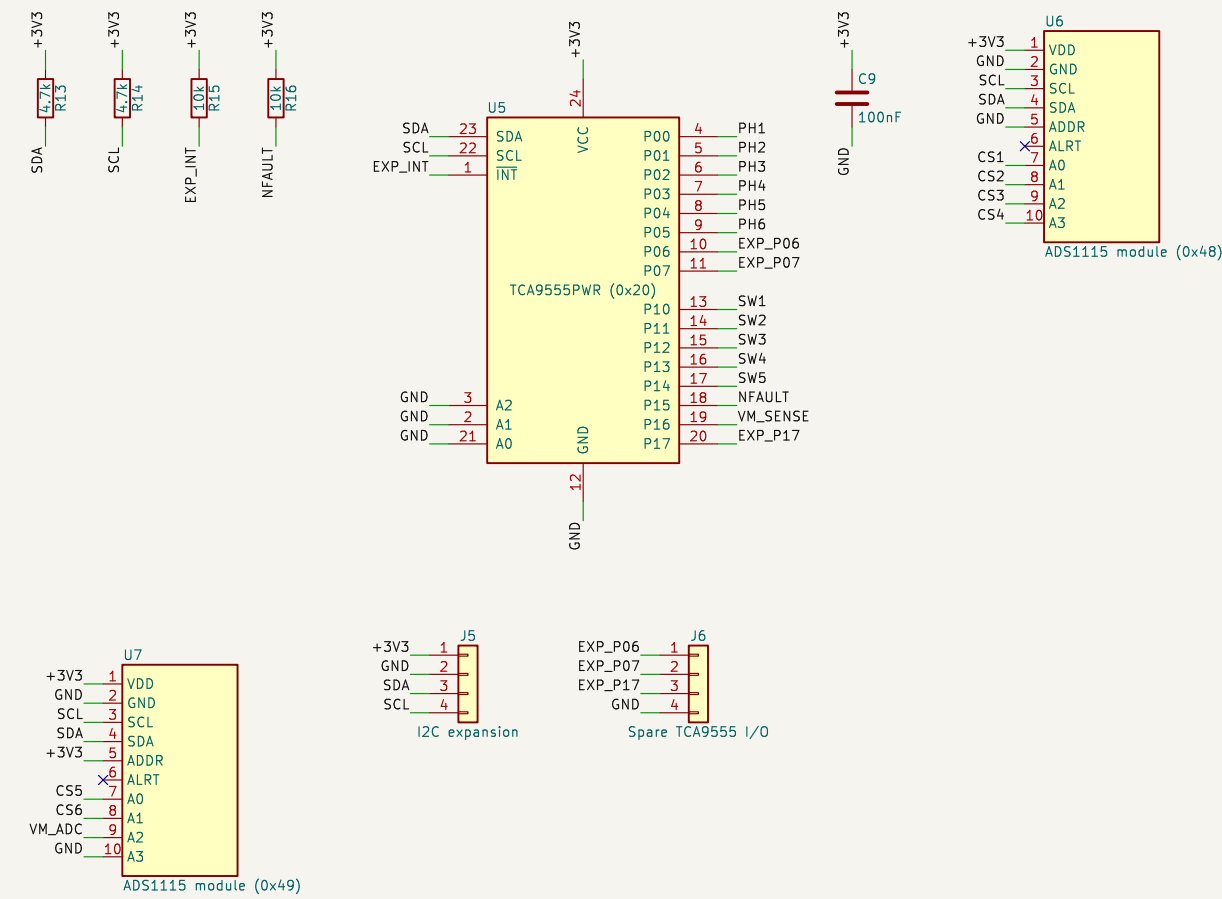
5 V BUCK + DEVKIT 5V FEED



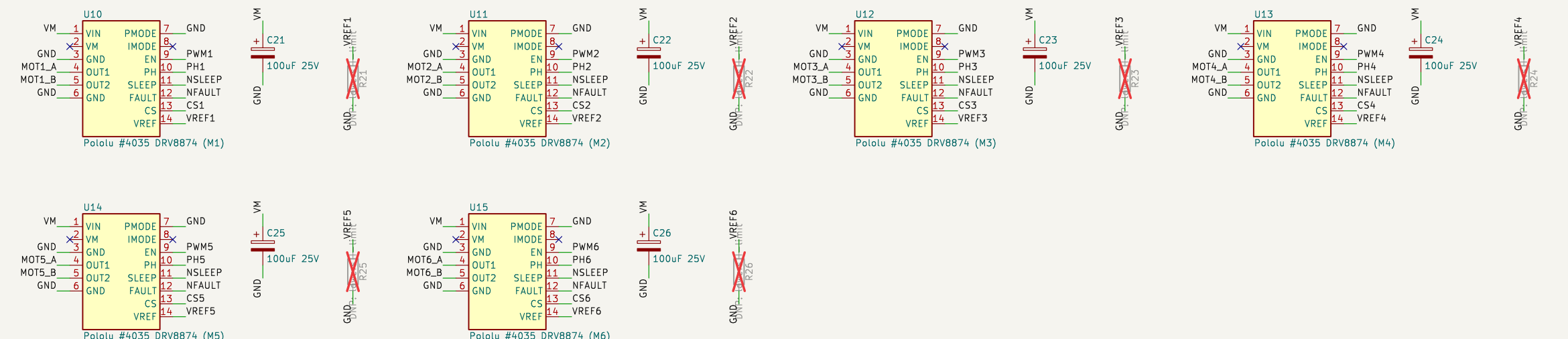
ESP32-DevKitC-32E (plug-in, 2x 1x19 female headers)



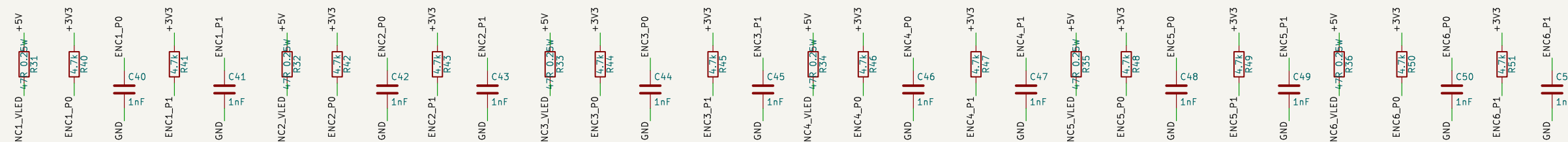
I2C: TCA9555 (0x20) + ADS7828 (0x48)



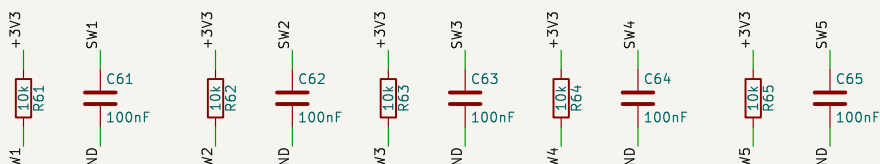
MOTOR DRIVERS: 6x Pololu DRV8874 carrier (plug-in) – PIN ORDER PLACEHOLDER



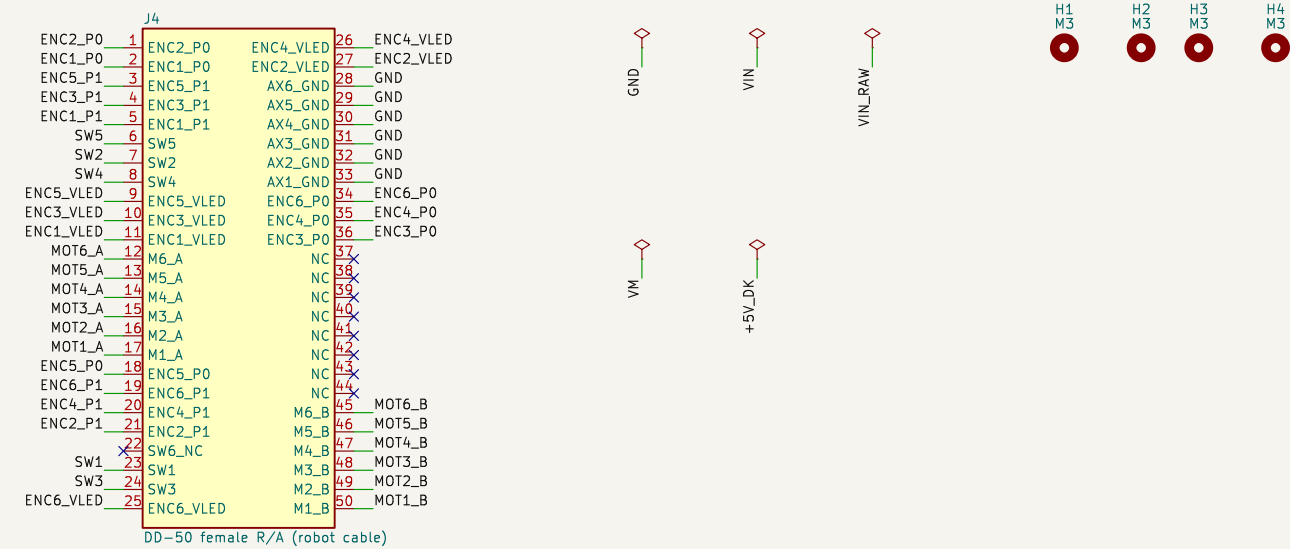
ENCODER FRONT END: 47R VLED, 10k pull-ups to 5V, 1nF, 74LVC14 at 3.3V



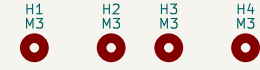
HOME SWITCHES (to TCA9555 P1.0–P1.4)



ROBOT CABLE: DD-50 female (SCORBOT-ER 4u manual POWER) FLAGS



MOUNTING



NOTES

- 1. Pololu DRV8874 carrier pin ORDER is a placeholder (names are correct). Fix design.CARRIER_ROWS and regenerate.
- 2. DB50 pinout from SCORBOT-ER 4u User Manual ch. 8 (same as ER III Table D-1). Pins 37-44 unused, 22 = gripper switch (no connection in arm).
- 3. 74HC14 symbol is used for the 74VLC14 (same pinout). Powered from +3V3; Inputs are 5 V tolerant.
- 4. Carrier current limit with SLEEP at 3.3 V is about 2.9 A (VREF pulled to SLEEP by 10k on the carrier, CS has 2.49k). R21-R26 (DNP) lower it.
- 5. AD57828 uses its internal 2.5 V reference: CS reads 1.12 V; CS, so readings clip above about 2.2 A.